OUTPUT :

Dataset Preview :

```
   Square_Footage  Num_Bedrooms  Num_Bathrooms  Year_Built  Lot_Size  \
0            1360             2              1        1981  0.599637
1            4272             3              3        2016  4.753014
2            3592             1              2        2016  3.634823
3             966             1              2        1977  2.730667
4            4926             2              1        1993  4.699073

   Garage_Size  Neighborhood_Quality   House_Price
0            0                     5  2.623829e+05
1            1                     6  9.852609e+05
2            0                     9  7.779774e+05
3            1                     8  2.296989e+05
4            0                     8  1.041741e+06
```

Dataset Info:

```
RangeIndex: 1000 entries, 0 to 999
Data columns (total 8 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Square_Footage        1000 non-null   int64
 1   Num_Bedrooms          1000 non-null   int64
 2   Num_Bathrooms         1000 non-null   int64
 3   Year_Built            1000 non-null   int64
 4   Lot_Size              1000 non-null   float64
 5   Garage_Size           1000 non-null   int64
 6   Neighborhood_Quality  1000 non-null   int64
 7   House_Price           1000 non-null   float64
dtypes: float64(2), int64(6)
memory usage: 62.6 KB
None
```

Missing Values:

```
Square_Footage          0
Num_Bedrooms            0
Num_Bathrooms           0
Year_Built              0
Lot_Size                0
Garage_Size             0
Neighborhood_Quality    0
House_Price             0
dtype: int64
```

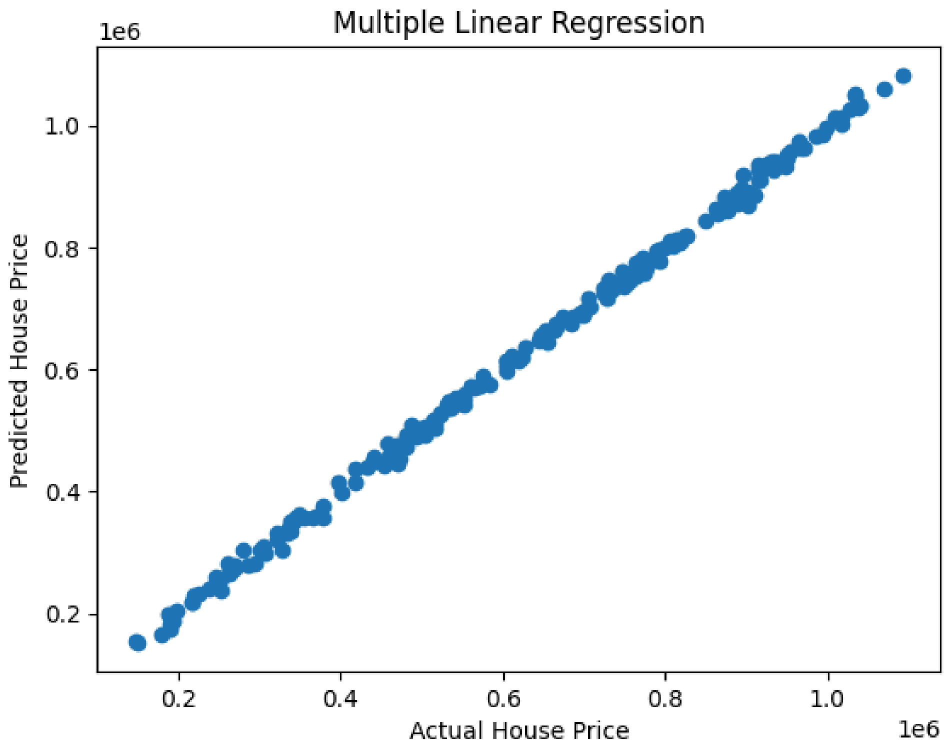
===== MULTIPLE LINEAR REGRESSION =====

MAE: 8174.583600006606

MSE: 101434798.50563568

RMSE: 10071.484424137074

R2 Score: 0.9984263636823413



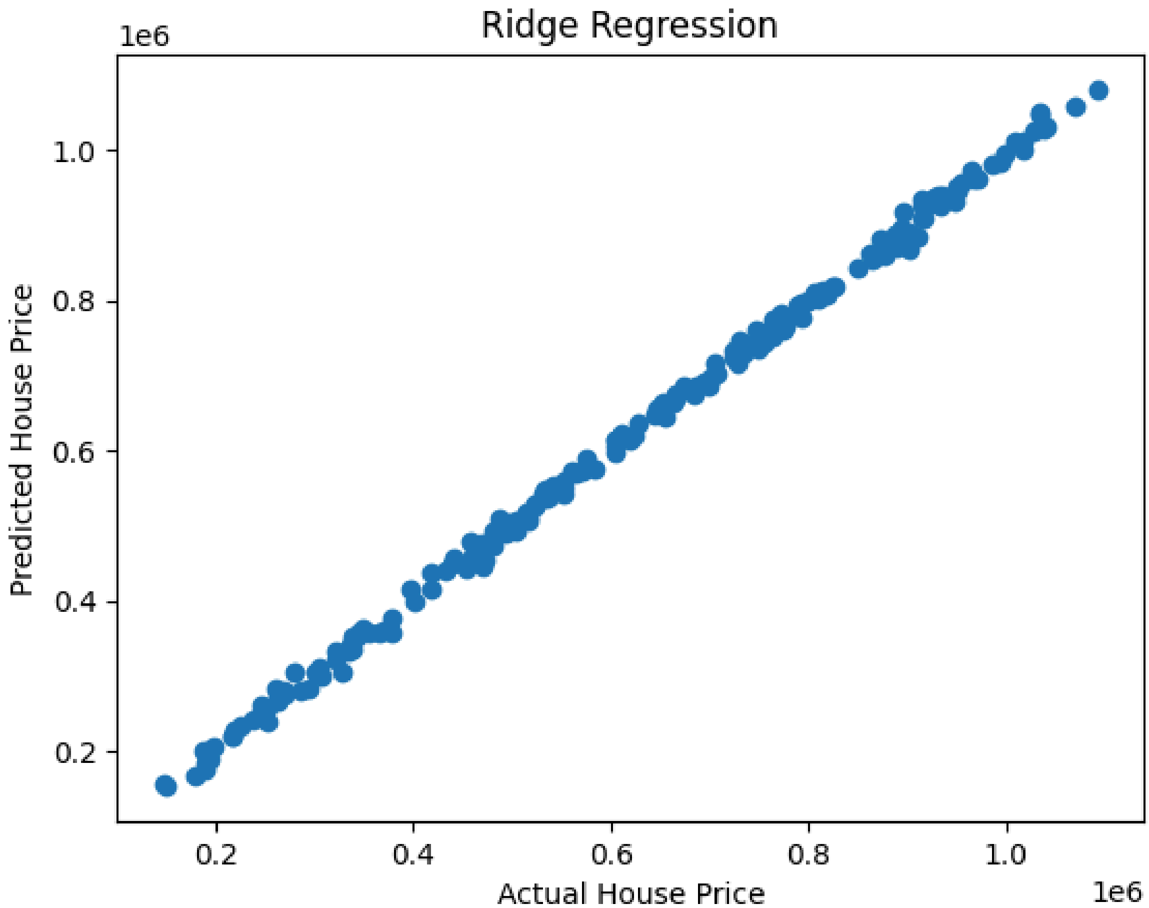
===== RIDGE REGRESSION =====

MAE: 8241.898358133865

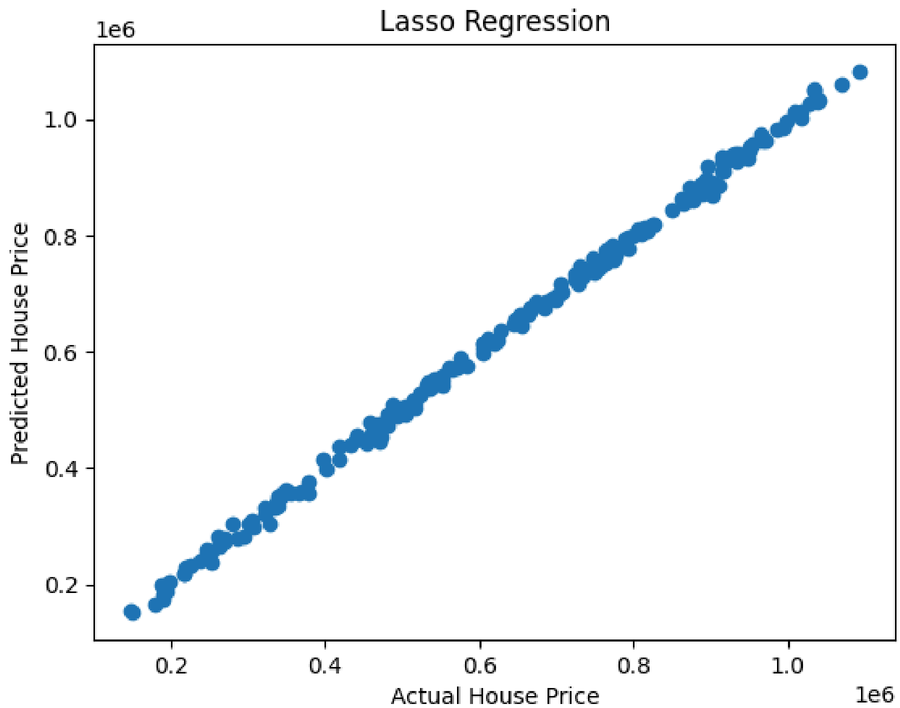
MSE: 102486578.62155962

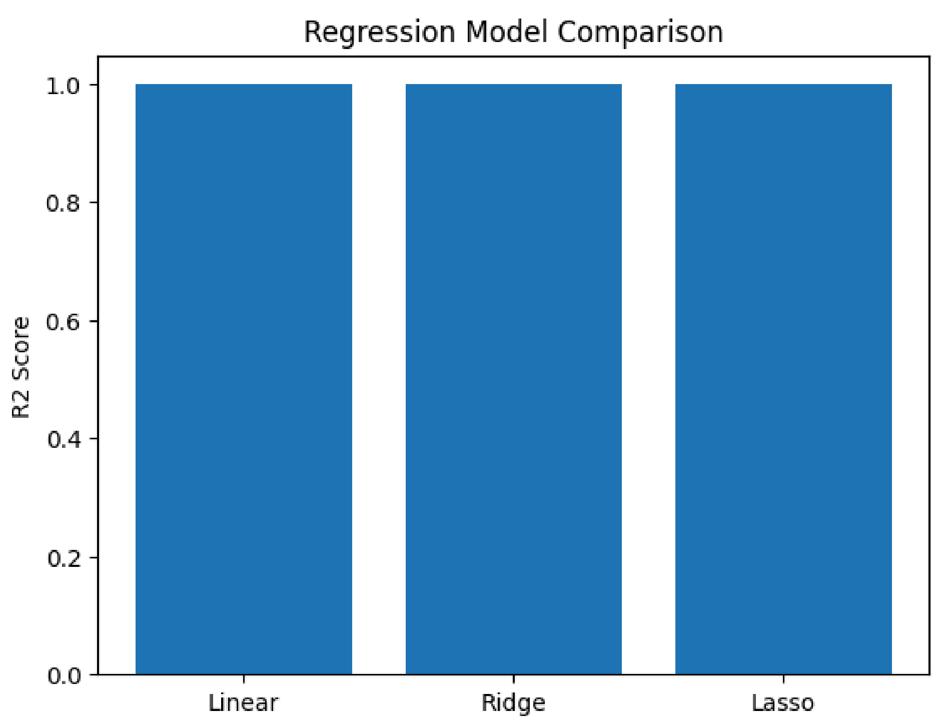
RMSE: 10123.565509323265

R2 Score: 0.998410046605628



===== LASSO REGRESSION =====
MAE: 8174.585809784338
MSE: 101434824.93285091
RMSE: 10071.48573611912
R2 Score: 0.9984263632723556





===== HYPERPARAMETER TUNING RESULTS =====

Best Ridge Alpha: {'alpha': 0.001}

Best Lasso Alpha: {'alpha': 10}